

Overview

Part 07 —  Agentic AI Systems · 07-agentic-ai/README.md

Module 07 — Agentic AI Systems

Where you are: Stop 7 of 16. **Time:** ~18–22 hours · **Prereq:** Module 06 — AI Agents [06-ai-agents] — non-negotiable.

Module 06 gave a single agent a loop, a set of tools, and a memory. This module asks the next question: what happens when one loop isn't enough — when you put two or more agents in a system together? Multi-agent systems are not "Module 06 but bigger." They are a different design problem: topology, communication, coordination failure, and a cost curve that punishes you for reaching for them by reflex. This module covers the theory of *how* agents should be arranged and *when* they should be arranged at all — the mechanics of any specific framework (LangGraph, CrewAI, AutoGen) live in Module 16 [16-ai-tech-stack].

Learning objectives

By the end of this module you can:

1. Place any system on the autonomy spectrum (single call → tool agent → multi-agent → autonomous long-runner) and argue for the *least* autonomous point that solves the problem.
2. Draw and evaluate the five core multi-agent topologies — supervisor, peer hand-off, hierarchical, debate/vote, pipeline — and name each one's specific failure mode from memory.
3. Quantify the token-cost multiplier of multi-agent systems and defend, with numbers, when it's worth paying.
4. Explain the three communication patterns (blackboard, message passing, shared state) and where A2A-style protocols sit relative to MCP.
5. Diagnose the five coordination failure modes unique to multi-agent systems (deadlock, information loss, hallucination poisoning, redundant work, debate thrashing) from a trace.
6. Choose a human-in-the-loop vs. human-on-the-loop oversight posture for a given autonomy level, with a concrete trigger condition.
7. Apply a decision framework to classify a real request as single-agent, multi-agent, or a deterministic pipeline with an LLM step — and defend the call.

Suggested path

#	Do this	File	Time
1	Autonomy spectrum + the five topologies	notes/01-autonomy-spectrum-and-topologies.md	4h
2	Cost, protocols, failure modes, control, evals	notes/02-coordination-control-and-evaluation.md	5h
3	Run and read the topology simulator	code/topology_simulator.py	2h
4	Run and read the hand-off information-loss demo	code/handoff_information_loss.py	1h
5	Slides	slides/	1h
6	Lab exercises	lab/EXERCISES.md	6h
7	Papers	papers/PAPERS.md	3h

The 18 terms you must own

autonomy spectrum · orchestrator-worker (supervisor) · peer hand-off / swarm · hierarchical teams-of-teams · debate / parallel-then-vote · pipeline (assembly-line) · context isolation · hand-off summarization loss · blackboard / scratchpad · structured message passing · shared state (channels) · A2A (Agent2Agent) · deadlock / livelock · hallucination poisoning · human-in-the-loop (HITL) · human-on-the-loop (HOTL) · goal drift · budget governor

Exit check

Given a real (or realistic) automation request, you can produce a one-page decision memo: where it sits on the autonomy spectrum, which topology (if any) fits the problem shape, the token-cost estimate versus a single-agent baseline, the two most likely coordination failure modes and how you'd detect them, and whether the system needs HITL or HOTL oversight — with a named trigger condition for escalation.